SANJAY SARELLA

DATA ANALYTICS | SQL | TABLEAU | POWER BI | PYTHON

sanjaysarella11@gmail.com | 405-861-0196 | LinkedIn | GitHub

SUMMARY

Data Analytics graduate with practical experience using business and operational data for reporting, forecasting, trend analysis, and decision support. Skilled in SQL, Excel, Tableau, Power BI, Python, pandas, data visualization, KPI analysis, and machine learning. Procurement experience includes budgeting, purchasing trends, supplier coordination, and management reporting, supported by deployed analytics projects.

SKILLS

Analytics & BI: SQL, Excel, Tableau, Power BI, exploratory data analysis, KPI analysis, trend analysis, dashboard design, forecasting, data cleaning, data storytelling

Programming & Data: Python, pandas, NumPy, SQLite, DuckDB, Jupyter, REST APIs, Git/GitHub

Statistics & Machine Learning: scikit-learn, XGBoost, Random Forest, classification, regression, feature engineering, time-series analysis, ROC-AUC, MAE/RMSE, SHAP

Additional Tools: FastAPI, LangGraph, Gemini/Vertex AI, ChromaDB, Docker, Google Cloud Run

EXPERIENCE

AI Project-Based Learning Experience | Easy Stones | Dallas, TX **Jan 2026 - May 2026**

- Participated in project-based learning focused on conversational and agent-based AI workflows that connect user questions with business data, retrieval, APIs, and controlled tool execution.
- Used Python, LangGraph, FastAPI, and vector retrieval concepts while supporting inventory- and pricing-oriented learning use cases.
- Practiced structured outputs, prompt design, testing, and cloud deployment while learning how technical outputs can be translated into understandable business workflows.

Procurement Manager | Vegrow Pvt Ltd | India **Jul 2023 - Oct 2024**

- Collaborated with finance teams on budgeting and forecasting activities, helping represent procurement expenses accurately and identify cost-saving opportunities.
- Managed the supply-chain process from order placement through delivery to support timely and accurate fulfillment.
- Prepared and presented management reports on purchasing trends and cost savings to support sourcing and operational decisions.
- Developed relationships with key suppliers that improved communication, pricing discussions, and delivery reliability.

PROJECTS

Agentic Excel Analyst - Workbook Intelligence Platform | Live Demo

Tools: Python | SQL | DuckDB | pandas | Excel | Gemini/Vertex AI | FastAPI | Google Cloud Run

- Built a conversational analytics application that inspects unfamiliar Excel workbooks, identifies sheets, measures, dimensions, dates, and candidate relationships, and answers business questions without a predefined schema.
- Used deterministic SQL and restricted Python/pandas analysis for aggregations, joins, rankings, ratios, temporal comparisons, statistics, and anomaly-oriented analysis.
- Added relationship validation for cross-sheet analysis so joins are used only when supported by discovered workbook structure.

• Implemented evidence capture and answer verification so analytical claims are tied back to executed calculations and source workbook context.

Retail Demand Forecasting & Inventory Decision Support | Live Demo | GitHub

Tools: Python | SQL/SQLite | pandas | XGBoost | Prophet | SHAP | Power BI | Streamlit

• Analyzed 421,570 weekly-sales records across 45 stores using pandas transformations, SQLite queries, and visual analysis of store, department, holiday, and sales patterns.

• Engineered 26 temporal, store, promotion, and economic features for XGBoost sales modeling, used SHAP artifacts to explain model behavior, and preserved time order with a chronological split.

• Joined sales with store and economic features, created Power BI reporting artifacts, and connected analytical outputs to a Streamlit decision-support workflow.

• Used SQLite queries to compare holiday effects, store types, departments, and sales ranges for business-oriented exploratory analysis.

Customer Churn Prediction & Retention Intelligence | Live Demo | GitHub | Tableau

Tools: Python | pandas | scikit-learn | XGBoost | Random Forest | SHAP | Tableau

• Built a telecom churn analysis application using 7,043 customer records, covering data preparation, exploratory analysis, model comparison, prediction, and customer-level feature explanations.

• Compared Random Forest and XGBoost on a stratified 80/20 split; the saved modeling notebook records ROC-AUC values of 0.8225 and 0.8185 respectively.

• Used SHAP explanations, a Tableau dashboard, and a deployed application to communicate churn patterns and retention-oriented findings.

• Prepared contract, tenure, service, demographic, and billing variables by converting numeric fields, handling missing values, and encoding categorical features.

Federal Contract Intelligence - Procurement Analytics | Live Demo | GitHub

Tools: Python | pandas | USASpending.gov API | Random Forest | SHAP | FastAPI | React

• Built a procurement analytics prototype from 969 USASpending.gov award records and engineered nine features covering award amount, agency activity, duration, geography, timing, and recipient history.

• Trained a Random Forest to classify definitive contracts versus delivery orders and evaluated 194 saved test predictions at 0.741 ROC-AUC, with SHAP explanations for model outputs.

• Integrated public-data ingestion, model inference, explanations, and procurement-oriented recommendations into an interactive React/FastAPI application.

ORION - Corporate Distress Intelligence | Live Demo | GitHub

Tools: Python | FastAPI | SEC EDGAR/XBRL | Financial Ratios | Async APIs | React | LangGraph

• Developed a financial-research prototype that combines SEC EDGAR company facts, Form 4 insider filings, news signals, and FRED macroeconomic context for company-risk screening and research.

• Implemented explainable financial indicators, historical trend views, and SIC-based peer comparisons, with up to five peers when relevant data is available.

• Generated four stakeholder-specific brief formats - Executive, Consultant, Investor, and Board - in a React/FastAPI application to communicate findings for different analytical audiences.

EDUCATION

Master of Science in Data Analytics | Oklahoma City University — Dec 2025

Bachelor of Science | Oriental University — May 2023